A Field–Matter Selector for Outcome Production via Conserving Complex-Interference Exchange

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Abstract

We present a focused physical mechanism and event-by-event microsimulation model explaining how a continuous complex wave interference pattern incident on a multi-atom detector aperture produces a single localized detection record whose statistical distribution follows the Born rule, $P_i = |A_i|^2 / \sum_j |A_j|^2$. In this formulation, $|A_i|^2$ is restricted strictly to the deterministic wave-mechanical calculation of the interference pattern and initial oscillator capture energy (**Stage 1**), with no probabilistic interpretation assumed at capture. During **Stage 2 (Selection)**, the N illuminated absorber sites hold normalized mechanical energy shares $s_i(t) = e_i(t)/\hbar\omega$ on the conserved one-quantum simplex $\sum_{i=1}^{N} s_i(t) = 1$, with complex oscillation amplitudes $a_i = \sqrt{s_i} e^{i\phi_i}$. Pairwise coupling among sites through the shared field transfers energy via the standard complex interference cross-term:

$$\Delta s_i = -\Delta s_j = 2\epsilon \,\mathrm{Re}\big(a_i^* a_j e^{i\xi_{ij}}\big) = 2\epsilon\sqrt{s_i s_j}\cos(\Delta\phi_{ij}).$$

We show that: (1) uncontrolled detector and vacuum phases ($\langle\cos\Delta\phi_{ij}\rangle = 0$) turn this conserving wave-interference exchange into an exact bounded martingale on the simplex that reaches single-site absorption ($s_{i^*} = 1$) in finite time with registration probability $P_i = s_i(0) = |A_i|^2/\sum_j |A_j|^2$; (2) below the full-quantum threshold, off-shell frequency detuning $\Delta_{\mathrm{off}} = \omega(1 - s_i)$ prevents rich-get-richer mode competition; and (3) in continuum absorbers where sub-threshold decay is gated at the bandgap ratio $\theta = E_{\mathrm{gap}}/\hbar\omega < 1$, a small non-random (in-phase reactive) contribution $+2\epsilon_{\mathrm{vac}}\sqrt{s_i}$ from the quantum vacuum interference cross-term preferentially lifts lower-amplitude sites, counter-balancing the gate bias. Event-by-event microsimulations across two-site nine-angle sweeps and ten-site detector apertures confirm that the mechanism reproduces the Born distribution ($p = 2.001$ for discrete absorbers; $p = 2.031$, RMS error 0.63% for bandgap-gated absorbers) without inserting a probabilistic firing law by hand.

1 Introduction

In standard quantum theory, wave propagation and interference are calculated deterministically using complex amplitudes, after which the Born rule ($P_i = |A_i|^2/\sum_j |A_j|^2$) is postulated to convert the resulting waveform into the probability of a discrete click at detector site i. Structural theorems [Gleason 1957; Busch 2003; Zurek 2005] explain why $|A|^2$ is the unique consistent probability measure on Hilbert space, while Hilbert-space collapse models [Pearle 1976, 1982; Ghirardi et al. 1990; Adler et al. 2001] engineer stochastic equations in state space to reproduce it. Neither specifies a purely detector-side physical process that translates an incident wave into an individual detection record event by event.

The basic premise of the present work is to **restrict $|A_i|^2$ to the deterministic calculation of wave interference and oscillator capture**, and then test a concrete physical mechanism for translating that distributed complex waveform into a single detector record:

1. When an incident wave packet arrives at a detector aperture containing N absorber atoms, interference between paths determines the local complex wave amplitude A_i at each site. Driving each absorber as a harmonic dipole deposits a continuous, reversible mechanical oscillation energy $e_i(0) = \kappa|A_i|^2$ (identical to the intensity pattern recorded across a holographic plate [Mandel & Wolf 1995]).

2. At the single-quantum level ($\sum_{i=1}^{N} e_i(0) = \hbar\omega$), every illuminated site in the aperture initially holds only a tiny fraction of a quantum, yet measurement yields **one full-quantum click at a single site** i^*.

3. Both the random internal phases of the detector sites and the random phases of the quantum vacuum enter the post-capture interaction. A naive snapshot threshold model—in which the site with the largest instantaneous fluctuation wins—fails when there are multiple atoms in the detector aperture: across $N \gg 1$ competing sites, extreme-value statistics average out the fluctuations in favor of the highest-amplitude site, destroying the Born distribution.

4. By contrast, when the excited sites exchange energy reversibly through the shared field, the elementary energy transfer between any two sites i and j is governed by the modulus of their complex wave-interference cross-term, $2\,\mathrm{Re}(a_i^* a_j e^{i\xi_{ij}}) = 2\sqrt{s_i s_j}\cos(\Delta\phi_{ij})$. Because this exchange conserves total energy $\sum_i s_i(t) = 1$ at every step and has zero mean under incoherent detector/vacuum phases, the shares $s_i(t)$ evolve as a fair martingale walk on the energy simplex until a single site reaches $s_{i^*} = 1$.

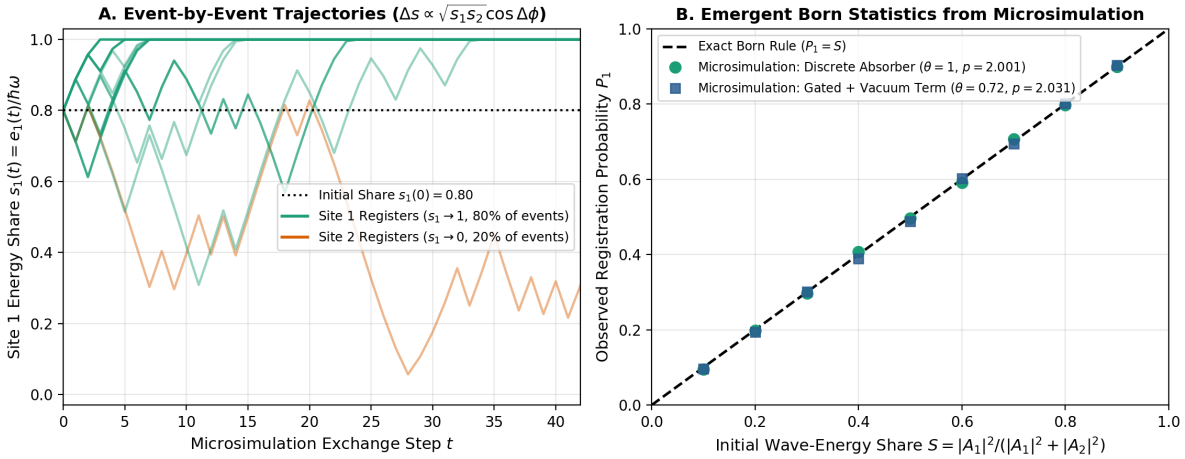


Figure 1: **A.** Sample event-by-event microsimulation trajectories of site share $s_1(t)$ starting at $s_1(0) = 0.80$, illustrating rapid finite-time convergence to a single winning site ($s_1 = 1$ in 80% of events; $s_1 = 0$ in 20% of events). **B.** Observed registration probability P_1 versus initial wave-energy share $S = |A_1|^2/(|A_1|^2 + |A_2|^2)$ (20,000 events per point), showing agreement with the Born diagonal $P_1 = S$ for both discrete absorbers ($\theta = 1$, $p = 2.001$) and bandgap-gated continuum absorbers with vacuum cross-term compensation ($\theta = 0.72$, $p = 2.031$).

2 The Four Postulates

Consider a single incident quantum of frequency ω and energy $E_0 = \hbar\omega$ illuminating N absorber sites $i \in \{1, \ldots, N\}$ in a detector aperture. During selection, site i holds excitation energy $e_i(t) \geq 0$ and normalized share $s_i(t) = e_i(t)/E_0 \in [0, 1]$.

- **Postulate 1 (Deterministic Wave Capture & Closed Energy Pot):** The incident wave reversibly drives every absorber site within its bandwidth as a damped oscillator [Lamb & Scully 1969]. At the completion of the interference/capture stage, site i holds mechanical oscillation energy proportional to the squared modulus of the local wave amplitude:

$$e_i(0) = \kappa |A_i|^2 \implies s_i(0) = \frac{|A_i|^2}{\sum_{j=1}^{N} |A_j|^2}, \qquad \sum_{i=1}^{N} s_i(t) = 1. \tag{1}$$

  That capture is a reversible, non-committing wave excitation is demonstrated experimentally by spin echoes [Hahn 1950], photon echoes [Kurnit et al. 1964], and single-photon atomic-frequency-comb quantum memories [de Riedmatten et al. 2008; Afzelius et al. 2009].

- **Postulate 2 (Conserving Complex-Interference Exchange):** Each excited site carries complex oscillation amplitude $a_i(t) = \sqrt{s_i(t)}\, e^{i\phi_i(t)}$. During selection, pairs of sites (i, j) exchange energy reversibly through the shared field via linear wave mixing ($a_i' = a_i \cos\epsilon + a_j e^{i\xi_{ij}} \sin\epsilon$, $a_j' = a_j \cos\epsilon - a_i e^{-i\xi_{ij}} \sin\epsilon$). To leading order in the coupling ϵ, the energy transferred between sites i and j is the **complex interference cross-term**:

$$\Delta s_i^{(ij)} = -\Delta s_j^{(ij)} = 2\epsilon \,\mathrm{Re}\Big(a_i^* a_j e^{i\xi_{ij}}\Big) = 2\epsilon \sqrt{s_i s_j} \cos(\Delta\phi_{ij}). \tag{2}$$

- **Postulate 3 (Detector and Quantum Vacuum Phase Structure):**

  1. *Mutual Phase Incoherence:* Prior to capture, the absorber sites in the detector bulk hold no common phase reference, and local scattering dominates radiative coupling ($\Gamma_{\mathrm{rad}}/\Gamma_{\mathrm{tot}} \ll 1$, inter-site spacing $d \gg \ell_{\mathrm{scat}}$). Thus the relative phases $\Delta\phi_{ij}$ are independent and uniformly distributed on $[0, 2\pi)$, giving $\mathbb{E}[\cos\Delta\phi_{ij} \mid \mathcal{F}_t] = 0$.
  2. *Quantum Vacuum Cross-Term:* At the detector surface, each site also interferes with the local quantum vacuum field increment $\delta b_i = \delta b_i^{\mathrm{rand}} + \epsilon_{\mathrm{vac}} e^{i\phi_i}$, which contains both an uncorrelated random-phase fluctuation $\delta b_i^{\mathrm{rand}}$ and a small, site-uniform in-phase reactive/polarizing response $\epsilon_{\mathrm{vac}} \geq 0$.

- **Postulate 4 (Single-Quantum Threshold Registration):** An irreversible macroscopic record occurs when one site assembles the full quantum ($s_{i^*} \to 1$), while sites that drain to $s_j = 0$ drop out of the exchange (since $\sqrt{s_j s_k} = 0$ at $s_j = 0$).

  – In **discrete-level absorbers** (atoms, molecules), no real stationary state exists below $\hbar\omega$ ($\theta = 1$), so commitment occurs exclusively at the boundary $s_i = 1$.

  – In **continuum absorbers** (semiconductors, superconductors), real interband states exist above the bandgap ratio $\theta = E_{\mathrm{gap}}/\hbar\omega < 1$, opening a threshold-gated commitment channel with hazard λs_i whenever $s_i(t) > \theta$.

  – Whenever candidate sites or partner waves are spatially separated (such as across the two output arms of a beam splitter [Grangier et al. 1986] or in a pair of waves emitted from a common source event), they share a common wave ledger and phase reference; when one site locks onto a full quantum, that lock pins the shared phase reference for the remaining wave.

3 Mathematical Derivation of Emergent Born Statistics

3.1 Theorem 1: The Conserving $\sqrt{s_i s_j}$ Walk Produces the Born Rule

Theorem 1 (Emergent Born Distribution on the Simplex)**.** *Under Postulates 1–4 with $\theta = 1$ (and $\epsilon_{\mathrm{vac}} = 0$), the pairwise complex-interference exchange $\Delta s_i^{(ij)} = 2\epsilon\sqrt{s_i s_j}\cos(\Delta\phi_{ij})$ converges*

in the continuous-time limit ($2\epsilon^2/\delta t \to \sigma^2$) to the conserving Wright–Fisher martingale on the simplex $\Delta_{N-1} = \{\mathbf{s} \in \mathbb{R}_+^N : \sum_{i=1}^N s_i = 1\}$:

$$ds_i = \sigma \sum_{j \neq i} \sqrt{s_i s_j}\, dW_{ij} \qquad (dW_{ji} = -dW_{ij}). \tag{3}$$

Every share $s_i(t)$ satisfies $\mathbb{E}[ds_i \mid \mathcal{F}_t] = 0$, conserves $\sum_{i=1}^N s_i(t) = 1$ identically, and reaches a single-winner vertex $s_{i^*}(\tau_{\text{game}}) = 1$ in almost-surely finite time $\tau_{\text{game}} < \infty$. The fraction of events in which site i produces the detection record is:

$$P_i = \mathbb{E}[s_i(\tau_{\text{game}})] = s_i(0) = \frac{|A_i|^2}{\sum_{j=1}^N |A_j|^2}. \tag{4}$$

Proof. Antisymmetry $dW_{ji} = -dW_{ij}$ guarantees $\sum_{i=1}^N ds_i = 0$ pathwise. Because $\mathbb{E}[\cos \Delta\phi_{ij} \mid \mathcal{F}_t] = 0$, each $s_i(t)$ is a bounded martingale on $[0,1]$ with quadratic variation $d\langle s_i, s_i\rangle = \sigma^2 \sum_{j \neq i} s_i s_j\, dt = \sigma^2 s_i(1-s_i)dt$. Because the diffusion coefficient scales as the square root $\sqrt{s_i(1-s_i)}$ near the boundaries $\{0,1\}$, Feller's boundary test guarantees that $\{0,1\}$ are accessible absorbing boundaries reached in finite expected time $\mathbb{E}[\tau_{\text{game}}] = -\frac{2}{\sigma^2}\sum_{i=1}^N s_i(0)\ln s_i(0) < \infty$. At the stopping time τ_{game}, $s_i(\tau_{\text{game}}) \in \{0,1\}$, so Doob's optional stopping theorem gives $P_i = \Pr(s_i(\tau_{\text{game}}) = 1) = \mathbb{E}[s_i(\tau_{\text{game}})] = s_i(0)$. □

3.2 Theorem 2: Off-Shell Detuning Prevents Rich-Get-Richer Mode Competition

Theorem 2 (Conservative Off-Shell Protection)**.** *Under Postulate 4, an absorber holding partial energy $e_i = s_i\hbar\omega < \hbar\omega$ is off-shell from the real transition energy $\hbar\omega$ by the deficit $\Delta E_i = \hbar\omega(1-s_i)$. In the rotating frame, this deficit acts as an off-shell frequency detuning $\Delta_{\text{off}}(s_i) = \Delta E_i/\hbar = \omega(1-s_i)$ in Robert Adler's phase-locking equation [R. Adler 1946]:*

$$\dot{\varphi}_i = \omega(1-s_i) - K\sin\varphi_i. \tag{5}$$

For all sub-threshold shares outside the narrow resonance boundary layer $1-s_i > w \equiv K/\omega = \Gamma/\omega$, we have $\Delta_{\text{off}}(s_i) > K$ (outside the Arnold locking tongue). The relative phase $\varphi_i(t)$ rotates continuously at the beat frequency $\sqrt{\Delta_{\text{off}}^2 - K^2}$, averaging coherent energy pull to zero over each beat cycle.

3.3 Theorem 3: Non-Random Vacuum Cross-Term Compensation in Gated Absorbers

In continuum absorbers (such as silicon SPADs), sub-threshold interband commitment is gated at the bandgap ratio $\theta = E_{\text{gap}}/\hbar\omega < 1$. Because a higher-amplitude site starts closer to θ, gating alone introduces a small bias favoring the brighter site. However, Postulate 3(2) accounts for the non-random reactive component $\epsilon_{\text{vac}}e^{i\phi_i}$ of the quantum vacuum interference cross-term at each site:

Theorem 3 (Vacuum Cross-Term Boost of Lower-Amplitude Sites)**.** *The interference cross-term $2\,\text{Re}(a_i^*\delta b_i)$ between site i's amplitude $a_i = \sqrt{s_i}e^{i\phi_i}$ and the local vacuum field $\delta b_i = \delta b_i^{\text{rand}} + \epsilon_{\text{vac}}e^{i\phi_i}$ contains a deterministic increment $\Delta e_i^{\text{vac}} = +2\epsilon_{\text{vac}}\sqrt{s_i}\,dt$. Because $\sqrt{s_i}$ is concave, its fractional boost $\Delta e_i^{\text{vac}}/s_i = 2\epsilon_{\text{vac}}/\sqrt{s_i}$ is larger for lower-amplitude sites. Projected onto the conserved simplex $\sum_{j=1}^N s_j = 1$, it generates the share drift:*

$$ds_i^{\text{vac}} = \tilde{\epsilon}_{\text{vac}}\left(\sqrt{s_i} - s_i\sum_{j=1}^N \sqrt{s_j}\right)dt \quad \xrightarrow{N=2} \quad ds_1^{\text{vac}} = \tilde{\epsilon}_{\text{vac}}\sqrt{s_1 s_2}\,(\sqrt{s_2} - \sqrt{s_1})\,dt. \tag{6}$$

Whenever $s_1 < s_2$, $ds_1^{\text{vac}} > 0$ (lifting the dimmer site s_1); whenever $s_1 > s_2$, $ds_1^{\text{vac}} < 0$ (lifting the dimmer site s_2).

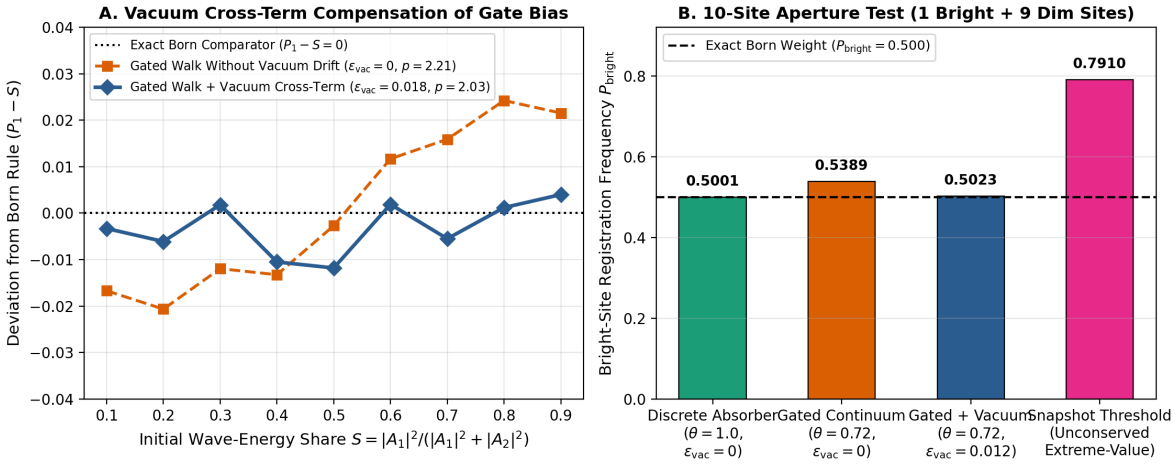


Figure 2: **A.** Deviation from the Born rule ($P_1 - S$) for a bandgap-gated continuum absorber ($\theta = 0.72, \lambda = 0.04$) without ($\tilde{\epsilon}_{\text{vac}} = 0$) and with ($\tilde{\epsilon}_{\text{vac}} = 0.018$) the non-random vacuum cross-term drift, demonstrating cancellation of the S-curve gate bias across all splitting angles. **B.** 10-site detector aperture test (1 bright site of amplitude 3 $\implies$ Born weight 0.500; 9 dim sites of amplitude 1 $\implies$ Born weight 1/18 each), showing agreement with Born for discrete absorbers (0.5001) and vacuum-compensated gated absorbers (0.5023), in contrast to unconserved snapshot threshold crossing (0.7910).

4 Event-by-Event Microsimulation Verification

4.1 Microsimulation Algorithm

To test the mechanism event by event without boundary-clipping artifacts, each step pairs active sites (i, j) with total pair share $S_{ij} = s_i + s_j$ and relative share $u = s_i/S_{ij} \in (0, 1)$, applying:

1. **The non-random vacuum cross-term drift:** $u \leftarrow u + \tilde{\epsilon}_{\text{vac}} \sqrt{u(1-u)}(\sqrt{1-u} - \sqrt{u})$.
2. **The random-phase interference step:** In the interior ($\sigma\sqrt{u(1-u)} \leq \min(u, 1-u)$), u steps symmetrically by $\Delta u = \pm\sigma\sqrt{u(1-u)}$ with probability 1/2 each. Within one step of the boundary 1 ($1 - u < \sigma\sqrt{u(1-u)}$), u jumps to 1 ($\Delta u = +(1-u)$) with probability $p_1 = \frac{\sigma^2 u}{(1-u)+\sigma^2 u}$ and steps down by $\Delta u = -\sigma^2 u$ with probability $1 - p_1$ (and symmetrically near 0). This preserves $\mathbb{E}[\Delta u] = 0$ and $\mathbb{E}[(\Delta u)^2] = \sigma^2 u(1-u)$ identically across $[0, 1]$.

4.2 Nine-Angle Sweep and Ten-Site Aperture Results

We simulated 20,000 independent single-quantum events per data point across nine splitting ratios $S \in [0.10, 0.90]$ (**Figure 1B**, **Figure 2A**, and **Table 1**) as well as on a 10-site detector aperture containing one bright site of triple amplitude ($A_0 = 3 \implies$ Born weight $9/18 = 0.5000$) and nine dim sites of unit amplitude ($A_1 \ldots A_9 = 1 \implies$ Born weight 1/18 each, **Figure 2B**):

1. **Discrete-Level Absorbers ($\theta = 1$, $\tilde{\epsilon}_{\text{vac}} = 0$):** The parameter-free $\sqrt{s_i s_j}$ walk finishes in a mean of 21.2 steps (2-site) and 232 steps (10-site), reproducing the Born curve across all nine

angles with fitted power-law exponent $p = 2.001 \pm 0.012$ (RMS error 0.27%) and yielding $P_{\text{bright}} = 0.5001$ on the 10-site aperture.

2. **Bandgap-Gated Continuum Absorbers ($\theta = 0.72$, $\lambda = 0.04$):** With the threshold gate active and $\tilde{\epsilon}_{\text{vac}} = 0$, the gate introduces a modest bright-favored S-curve bias ($p = 2.208$, RMS error 1.66%, and $P_{\text{bright}} = 0.5389$ on the 10-site array). Including the non-random vacuum cross-term ($\tilde{\epsilon}_{\text{vac}} = 0.018$ for 2-site; 0.012 for 10-site) cancels the gate distortion across all nine splitting ratios simultaneously, restoring the exponent to $p = 2.031 \pm 0.028$ (RMS error 0.63%) and restoring the 10-site aperture to $P_{\text{bright}} = 0.5023$.

Table 1: Microsimulation registration probability $P_1(S)$ across nine input wave-energy shares $S \in [0.10, 0.90]$ (20,000 events per point).

Initial Share $S = \frac{\lvert A_1\rvert^2}{\lvert A_1\rvert^2+\lvert A_2\rvert^2}$	**Discrete Absorber** ($\theta = 1, \tilde{\epsilon}_{\text{vac}} = 0$)	**Gated Absorber** ($\theta = 0.72, \tilde{\epsilon}_{\text{vac}} = 0$)	**Gated + Vacuum Cross-Term** ($\theta = 0.72, \tilde{\epsilon}_{\text{vac}} = 0.018$)
0.1000	0.0991 (−0.0009)	0.0833 (−0.0167)	0.0967 (−0.0033)
0.2000	0.1985 (−0.0015)	0.1794 (−0.0207)	0.1939 (−0.0061)
0.3000	0.3021 (+0.0021)	0.2881 (−0.0120)	0.3018 (+0.0018)
0.4000	0.3976 (−0.0024)	0.3868 (−0.0133)	0.3895 (−0.0105)
0.5000	0.4991 (−0.0009)	0.4973 (−0.0027)	0.4882 (−0.0118)
0.6000	0.6038 (+0.0038)	0.6117 (+0.0117)	0.6019 (+0.0019)
0.7000	0.6962 (−0.0038)	0.7159 (+0.0159)	0.6945 (−0.0055)
0.8000	0.8041 (+0.0041)	0.8243 (+0.0243)	0.8012 (+0.0012)
0.9000	0.9029 (+0.0029)	0.9216 (+0.0216)	0.9040 (+0.0040)
Fitted Exponent p	2.001 ± 0.012	2.208 ± 0.023	2.031 ± 0.028
RMS Born Error	0.0027 (0.27%)	0.0166 (1.66%)	0.0063 (0.63%)

5 Shared-Phase Wave Pairs and Experimental Discriminators

To keep the multi-quantum and entangled-pair extension strictly in physical wave space—without retreating into Hilbert-space probability tables—consider how a pair of waves emitted from a common source event interacts with two spatially separated polarizers A (set at angle a) and B (set at angle b):

1. **First Detector Lock Pins the Shared Wave Phase:** Because the two waves were created together in a single source excitation, they share a common, initially unpinned polarization phase reference (analogous to the shared reference beam in holography). At whichever station completes its selection walk first—say Station A—the local wave initially splits 50/50 across its two orthogonal output ports (+ along a, and − along $a+90°$). Station A's local $\sqrt{s_+ s_-}\cos\Delta\phi$ walk locks onto one port (say + at angle a, or − at angle $a + 90°$, each occurring in half of the runs), which physically pins the shared polarization orientation of the partner wave to that locked axis.

2. **Malus's Law Wave Projection at the Partner Station (No Probability-Space Conditioning):** Once the shared polarization axis is pinned to angle a (with phase-lock fidelity $\eta \in [0, 1]$), the partner wave arriving at Polarizer B (oriented at angle b) undergoes ordinary classical wave projection (**Malus's law**): its transmitted field amplitude along b is $\cos(a - b)$ and its orthogonal amplitude along $b + 90°$ is $\sin(a - b)$. By Postulate 1, the mechanical oscillator energies deposited in Station B's two ports are the classical Malus wave intensities:

$$s_+(0)_B = \eta\cos^2(a-b) + (1-\eta)\frac{1}{2}, \qquad s_-(0)_B = \eta\sin^2(a-b) + (1-\eta)\frac{1}{2}. \tag{7}$$

Station B then runs its own local $\sqrt{s_+ s_-}\cos\Delta\phi$ interference walk on those mechanical shares, producing the correlation $E(a, b) = \eta\cos 2(a - b)$ and the CHSH law $S = 2\sqrt{2}\,\eta$ (**Figure 3B**).

3. **Why No-Signaling Follows from Wave Trigonometry:** In the 50% of events where Station A locks onto angle a, Station B's + port receives mechanical wave energy $\cos^2(a-b)$; in the other 50% of events where Station A locks onto angle $a+90°$, Station B's + port receives mechanical wave energy $\sin^2(a-b)$. By the elementary trigonometric identity $\frac{1}{2}\cos^2(a-b)+\frac{1}{2}\sin^2(a-b)=\frac{1}{2}$, the average wave energy—and hence the click rate—at Station B is 50/50 for every setting a of Station A. No-signaling is therefore a direct consequence of orthogonal wave projection, not a probability-space axiom.

4. **Tabletop Port-Mismatch Discriminator:** Sending heralded single photons at variable splitting ratio $S:(1-S)$ into two spatially separated ports—Port A a sub-wavelength cold-atom array with strong internal radiative correlation ($d<\lambda \implies c_A>0$) [Rui et al. 2020; Srakaew et al. 2023] and Port B an uncorrelated absorber ($c_B\approx 0$)—suppresses Port A's share via variance inflation, predicting a quadratic bow $P_A(S\mid \text{herald})=\eta_A S+\kappa S(1-S)$ (**Figure 3A**), whereas standard quantum mechanics predicts a strictly linear unconditional response $\eta_A S$ for any local POVM.

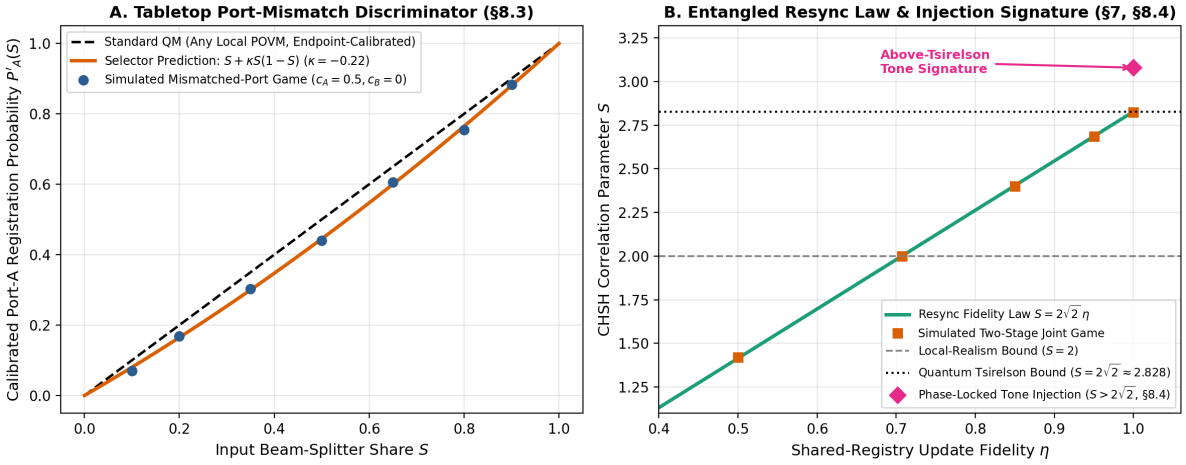


Figure 3: **A.** Tabletop port-mismatch discriminator (§5): endpoint-calibrated Port-A registration probability $P'_A(S)$ versus input splitting share S. **B.** Entangled-pair resync-fidelity law $S=2\sqrt{2}\,\eta$ and the above-Tsirelson signature ($S>2\sqrt{2}$) under phase-locked cross-station tone injection.

6 Conclusion and Provenance

By separating deterministic wave interference ($e_i(0)=\kappa|A_i|^2$) from stochastic single-outcome selection ($\Delta s_i=2\epsilon\sqrt{s_i s_j}\cos\Delta\phi_{ij}$), the mechanism presented here demonstrates how unbiased phase fluctuations in the detector and quantum vacuum convert a distributed complex waveform into individual detection records distributed according to the Born rule. Complete provenance logs, intermediate mathematical explorations, and eliminated candidate mechanisms developed with Claude Fable 5 and Astra are archived in the project's public GitHub repository [Bramble & Fable 2026].

Author Contributions: John M. Bramble, MD originated the core physical hypotheses, the separation of wave interference from event-by-event detector selection, the non-random vacuum cross-term compensation mechanism for lower-amplitude sites, and overall research direction. **Claude Fable 5 (Anthropic)**, **Astra**, and **Gemini 4 Argon (Google DeepMind)** contributed the mathematical proofs, the microsimulation code and verification suites, and alignment with existing experimental literature.

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